Problem 1)
$$\begin{cases} m v_i + M V_i = m v_f + M V_f \\ \frac{1}{2} m v_i^2 + \frac{1}{2} M V_i^2 = \frac{1}{2} m v_f^2 + \frac{1}{2} M V_f^2 \end{cases}$$

$$\Rightarrow \begin{cases} m(v_i - v_f) = M(V_f - V_i) \\ m(v_i^2 - v_f^2) = M(V_f^2 - V_i^2) \end{cases}$$
divide the two eqns.

$$\Rightarrow v_i + v_f = V_i + V_f \Rightarrow \boxed{v_i - V_i = -(v_f - V_f)}$$

Problem 2) a) $F = \frac{dP}{dt} = \frac{d(mv)}{dt} = m\frac{dv}{dt} + \frac{dm}{dt}v = 0 + \sigma v$

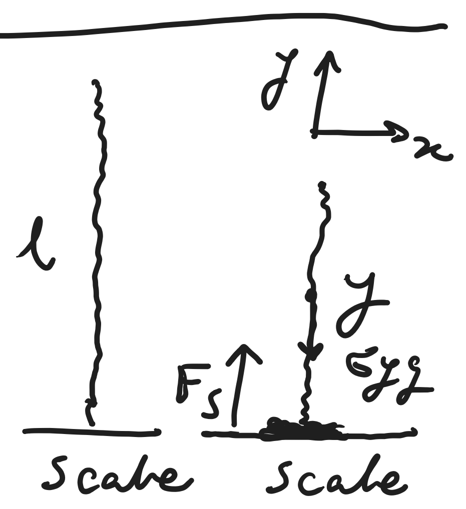
$\Rightarrow \boxed{F = \sigma v}$

b) $\frac{dT}{dt} = \frac{d}{dt}\left(\frac{1}{2} m v^2\right) = \frac{dm}{dt}\left(\frac{v^2}{2}\right) = \frac{\sigma v^2}{2}$

c) $\frac{dW}{dt} = \frac{F\,dx}{dt} = F.v = \sigma v^2$

d) Work-energy theorem states that, heat $= W - T$
$$= \sigma v^2 - \frac{\sigma v^2}{2} = \boxed{\frac{\sigma v^2}{2}}$$

Problem 3) $\sum F = F_s - \sigma L g = \frac{dP}{dt} = \frac{d(\sigma y \dot{y})}{dt}$
$$= \sigma y \ddot{y} + \sigma \dot{y}^2 \quad \text{(I)}$$

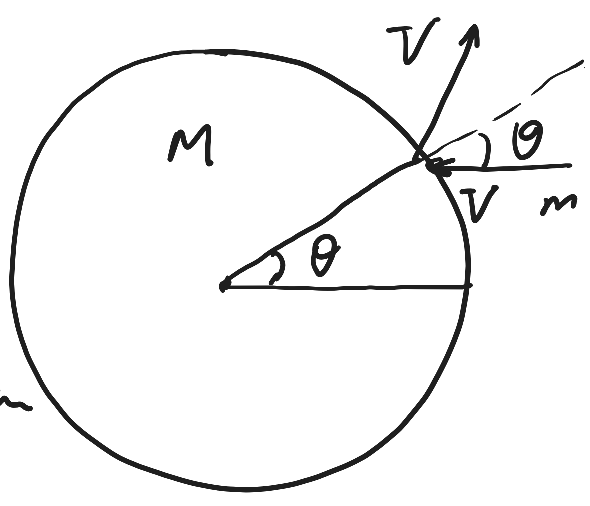


The part of chain in air is in free fall. $\Rightarrow \ddot{y} = -g$ (II)

From energy conservation we have, $\dot{y} = \sqrt{2g(L-y)}$ (III)

(I) (II) (III) $\Rightarrow F = \sigma L g - \sigma y g + 2\sigma(L-y)g = 3\sigma(L-y)g$

Problem 4) The horizontal velocity of m before and after collision is $-V$ and $V\cos 2\theta$. Thus, the net gain in momentum of m is $mV(1+\cos 2\theta)$.



Now, we need to compute the at which this momentum is transfered to compute $\vec{F} = \frac{d\vec{P}}{dt}$ the drag force.

The ring area at angle θ is,
$$dA = (2\pi R \sin\theta)(R\,d\theta)$$
However, only the effective (orthogonal area to V) is relevant to computing the rate of collisions. Thus,
$$dA_{eff} = dA \cos\theta$$
Thus, the volume swept by M at speed V is,
$$dv = V dA_{eff} = V(2\pi R \sin\theta)(R\,d\theta)\cos\theta$$
The collision rate is,
$$dp = n\,dv$$
The differential drag force is,
$$dF(\theta) = \frac{dP}{dt} = mV(1+\cos 2\theta)\,dp$$

$$\int_0^{\pi/2} dF(\theta) = \bar{F} = \int_0^{\pi/2} n(2V\pi R^2 \sin\theta\cos\theta)\, mV(1+\cos 2\theta)\,d\theta$$
$$= 2\pi n m R^2 V^2 \int_0^{\pi/2} \sin\theta\cos\theta(1+\cos 2\theta)\,d\theta$$
$$= \;〃\; \int_0^{\pi/2}\left(\frac{\sin 2\theta}{2} + \frac{\sin 4\theta}{4}\right)d\theta$$
$$= \;〃\; \left(-\frac{\cos 2\theta}{4} - \frac{\cos 4\theta}{16}\right)\Big|_0^{\pi/2}$$

$$\boxed{\bar{F} = \pi n m R^2 V^2}$$